Arista Access Node WFD-ACC-14 Whitefield POP Routing Protocol Flap SOP

Node: WFD-ACC-14 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5554

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a Routing Protocol Flap failure on the Arista Access Node WFD-ACC-14 located at the Whitefield POP. The failure domain involves BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. The root cause of this failure is attributed to the dynamic nature of routing protocols, which can lead to oscillations and flapping of routes. On Arista equipment, this can be exacerbated by the interaction between BGP and OSPF/ISIS, where route updates and topology changes can trigger flapping. Furthermore, the Access role on this node, responsible for connecting to the network core, is more susceptible to routing protocol flapping due to its critical position in the network topology. The severity of this failure is classified as minor, indicating that it does not impact network availability but may cause temporary packet loss and increased CPU utilization.

Pre-requisite Checks

- Step: Verify Physical Connection, Description: Check the physical connection between the Access Node WFD-ACC-14 and the network core to ensure it is secure and functioning correctly.

- Step: Verify Redundant Link Status, Description: Verify the status of the redundant link connected to the Access Node WFD-ACC-14 to ensure it is up and functioning correctly.

- Step: Dispatch Remote Hands, Description: Dispatch remote hands to the Whitefield POP to facilitate troubleshooting and repairs if necessary.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 10, main routing instance, Description: This command displays a summary of the BGP table, including the version number, main routing instance, and the number of BGP peers.

Command: show ip ospf database, Expected Output: OSPF database has 10 entries, Description: This command displays the OSPF database, including the number of entries and the types of entries present.
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```
Command: show isis database, Expected Output: ISIS database has 5 entries, Description: This command displays the ISIS database, including the number of entries and the types of entries present.
```

Resolution Steps

Step 1: Step: Clear BGP Flap Statistics, Command: clear bgp flap-statistics, Description: This command clears the BGP flap statistics, which can help to resolve flapping caused by BGP route churn.

Step 2: Step: Reset OSPF Adjacency, Command: reset ospf adjacency, Description: This command resets the OSPF adjacency, which can help to resolve flapping caused by OSPF adjacency loss.

Step 3: Step: Clear ISIS LSP Flood, Command: clear isis lsp flood, Description: This command clears the ISIS LSP flood, which can help to resolve flapping caused by ISIS LSP flooding.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 10, main routing instance, Description: This command verifies that the BGP table has stabilized and is functioning correctly.

Command: show ip ospf database, Expected Output: OSPF database has 10 entries, Description: This command verifies that the OSPF database has stabilized and is functioning correctly.

Command: show isis database, Expected Output: ISIS database has 5 entries, Description: This command verifies that the ISIS database has stabilized and is functioning correctly.
```

Roll-back Procedures

- Step: Restore Previous Configuration, Command: show running-config | include bgp|ospf|isis, Description: This command restores the previous configuration, including the BGP, OSPF, and ISIS configurations.

- Step: Reset BGP Flap Statistics, Command: clear bgp flap-statistics, Description: This command resets the BGP flap statistics to their previous state.

- Step: Reset OSPF Adjacency, Command: reset ospf adjacency, Description: This command resets the OSPF adjacency to its previous state.

- Step: Clear ISIS LSP Flood, Command: clear isis lsp flood, Description: This command clears the ISIS LSP flood to its previous state.